

**Subject Description Form**

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| <b>Subject Code</b>                              | ITC6983RT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Subject Title</b>                             | Guided Study in Advanced Materials and Structures: Auxetic Textiles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Credit Value</b>                              | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Level</b>                                     | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Pre-requisite / Co-requisite / Exclusion</b>  | Nil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Objectives</b>                                | This subject provides a guidance for research students to carry out a systematic study of auxetic textiles and to explore applications of such non-conventional textile materials in functional clothing and technical areas.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Intended Learning Outcomes</b>                | <p>Upon completion of the subject, students will be able to:</p> <ul style="list-style-type: none"><li>a. Comprehend basic concepts of auxetic materials and their nonconventional behaviors</li><li>b. Catch the up-to-date information about different types of auxetic materials and their applications</li><li>c. Apply different auxetic geometries to develop auxetic materials by using textile technology</li><li>d. Explore applications of auxetic textiles in different areas</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b> | <ul style="list-style-type: none"><li>a. Basic concepts of auxetic materials<br/>Definition of Poisson's ratio, effect of Poisson's ration on materials properties, auxetic and non-auxetic behaviors.</li><li>b. Auxetic geometries<br/>Reentrant geometries, rotating units, folded structures, chiral structures, star honeycomb structures.</li><li>c. Auxetic fibers and yarns<br/>Auxetic fibers formed based on special macromolecular structure and special spinning process, auxetic double helical yarns, auxetic composite yarns.</li><li>d. Auxetic fabrics<br/>Methods used to produce auxetic fabrics, auxetic knitted fabrics, auxetic woven fabrics, auxetic braided fabrics, auxetic non-woven fabrics.</li><li>e. Auxetic composites<br/>Methods used to produce auxetic composites, auxetic laminated composites, auxetic textile composites.</li><li>f. Applications of auxetic textile materials<br/>Applications of auxetic textiles in clothing and technical areas, such as high performance underwear, sportswear, protective clothing, medical care, etc.</li></ul> |

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| Teaching/Learning Methodology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | This is a guided study course and will have derived self-studying. The subject lecturer will suggest reading materials based on student’s research area and discuss with the students. Students are required to attend lectures, read assigned research papers, and submit a written report on an assigned research topic. |             |                                                                                |   |          |   |  |  |
| Assessment Methods in Alignment with Intended Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Specific assessment methods/tasks                                                                                                                                                                                                                                                                                          | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |   |          |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                            |             | a                                                                              | b | c        | d |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1. Coursework                                                                                                                                                                                                                                                                                                              | 40%         | ✓                                                                              | ✓ | ✓        | ✓ |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2. Examination                                                                                                                                                                                                                                                                                                             | 60%         | ✓                                                                              | ✓ | ✓        | ✓ |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Total                                                                                                                                                                                                                                                                                                                      | 100 %       |                                                                                |   |          |   |  |  |
| Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:<br><br>Coursework provides an opportunity to demonstrate the learning outcomes of students and to interact between the students and the lecturer, so that the understanding of the subject matter can be ensured.<br><br>Examination can assess students’ critical thinking and scientific analysis of various methods used in auxetic textile research.<br><br>The materials submitted for this assessment must be the student’s own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting GenAI-generated materials as students’ own work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions. |                                                                                                                                                                                                                                                                                                                            |             |                                                                                |   |          |   |  |  |
| Student Study Effort Expected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Class contact:                                                                                                                                                                                                                                                                                                             |             |                                                                                |   |          |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ▪ Tutorial/presentation/discussion                                                                                                                                                                                                                                                                                         |             |                                                                                |   | 24 Hrs.  |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ▪ Lab demonstration                                                                                                                                                                                                                                                                                                        |             |                                                                                |   | 2 Hrs.   |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Other student study effort:                                                                                                                                                                                                                                                                                                |             |                                                                                |   |          |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ▪ Self-reading                                                                                                                                                                                                                                                                                                             |             |                                                                                |   | 50 Hrs.  |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ▪ Assignment and preparation of report                                                                                                                                                                                                                                                                                     |             |                                                                                |   | 50 Hrs.  |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Total student study effort                                                                                                                                                                                                                                                                                                 |             |                                                                                |   | 126 Hrs. |   |  |  |
| Reading List and References                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <u>Books:</u>                                                                                                                                                                                                                                                                                                              |             |                                                                                |   |          |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1. Hong Hu, Minglonghai Zhang, Yanping Liu, <i>Auxetic Textiles</i> , ISBN: 978-0-08-102211-5, Elsevier, 2019.                                                                                                                                                                                                             |             |                                                                                |   |          |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2. Teik-Cheng Lim. <i>Auxetic Materials and Structures</i> . Springer, 2015.                                                                                                                                                                                                                                               |             |                                                                                |   |          |   |  |  |
| <u>Journals:</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                            |             |                                                                                |   |          |   |  |  |
| 1. Muhammad Zeeshan, Hong Hu and Adeel Zulifqar, Three-dimensional narrow woven fabric with in-plane auxetic behavior, <i>Textile Research Journal</i> , DOI: 10.1177/00405175221109639                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                            |             |                                                                                |   |          |   |  |  |
| 2. Chang Yuping, Yanping Liu, and Hu Hong, Deformation Behavior of Auxetic Laminated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                            |             |                                                                                |   |          |   |  |  |

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|  | <p>Fabrics with Rotating Square Geometry, Textile Research Journal, DOI: 10.1177/00405175221109629</p> <ol style="list-style-type: none"> <li>3. Yuping Chang and Hong Hu*, 3D Fabrics with Negative Poisson's Ratio: A Review, Applied Composite Materials, doi.org/10.1007/s10443-021-09931-w</li> <li>4. Peng Fan, Yu Chen, Jian Xiong and Hong Hu*, In-plane mechanical properties of a novel hybrid auxetic structure, Smart Materials and Structures, 2022, 31, 075003.</li> <li>5. Zhengkai Zhang, Qingguo Wen, Pengju Li, Hong Hu. Application of double arrowhead auxetic honeycomb structure in displacement measurement, Sensors and Actuators: A. Physical, 2022, 333, 113218.</li> <li>6. Yuping Chang, Yanping Liu, and Hong Hu*, Design and manufacture of 3D auxetic warp knitted spacer fabrics based on re-entrant and rotating geometries, Textile Research Journal, 2022, 92(3-4), 467-478. DOI: 10.1177/00405175211037204.</li> <li>7. Minglonghai Zhang, Hong Hu*, Hasan Kamrul, Shuaiquan Zhao, Yuping Chang, Mabel Ho, Nazmul Karim, Three-dimensional Composites with Nearly Isotropic Negative Poisson's Ratio by Random Inclusions: Experiments and Finite Element Simulation, Composite Science and Technology, 2022, 218, 109195.</li> <li>8. Vladislav A. Lvov et al. Auxetic Metamaterials for Biomedical Devices: Current Situation, Main Challenges, and Research Trends, Polymers, 2022, 15, 1439.</li> <li>9. Yu Chen, Ming-Hui Fu*, Hong Hu*, Jian Xiong, Curved inserts in auxetic honeycomb for property enhancement and design flexibility, Composite Structures, 2022, 280, 114892 doi.org/10.1016/j.compstruct.2021.114892</li> <li>10. Qingguo Wen, Pengju Li, Zhengkai Zhang, Hong Hu. Application of negative Poisson's ratio honeycomb structure in force measurement, Sensor Review, 2021, 42(1), 1-7.</li> <li>11. Yujin Kim, Kuk Hui Son and Jin Woo Lee, Auxetic Structures for Tissue Engineering Scaffolds and Biomedical Devices, Polymers, 2021,14, 6821.</li> <li>12. Andrew Alderson. A triumph of lateral thought, Chemistry &amp; Industry, 17 May 2019, 384-391.</li> <li>13. M.N. Ali, I.U. Rehman, Auxetic polyurethane stents and stent-grafts for the palliative treatment of squamous cell carcinomas of the proximal and mid oesophagus: A novel fabrication route, J. Manuf. Syst. 37 (2015) 375–395.</li> <li>14. M.N. Ali, I.U. Rehman, An Auxetic structure configured as oesophageal stent with potential to be used for palliative treatment of oesophageal cancer; Development and in vitro mechanical analysis, J. Mater. Sci. Mater. Med. 22 (2011) 2573–2581.</li> <li>15. Lee J. Hall, Vitor R. Coluci, Douglas S. Galvão, Mikhail E. Kozlov, Mei Zhang, Sócrates O. Dantas, Ray H. Baughman, Sign Change of Poisson's Ratio for Carbon Nanotube Sheets, Science 320, 504-507 (2008)</li> <li>16. Ray H. Baughman et al. Negative Poisson's ratios as a common feature of cubic metals, NATURE 1998, 392 (26): 362-265.</li> </ol> |
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